

You have successfully created 12 subnets: subnet-0ea60349aceff7f6e, subnet-0ff59b0fe30bdc66, subnet-08eb40900ab4e7203, subnet-02f8e2a7d7e3f9956, subnet-056f1b3a272015364, subnet-03e84f41c7b591bff, subnet-07ac5132e6aadbc3, subnet-07d4d875c492909ba, subnet-099e3d6e789cfcdb1, subnet-038bdc2207790e850, subnet-0e3f29dea7c33bfab, subnet-08d9add449a889471

Subnets (15) Info

Last updated 1 minute ago

Actions

Create subnet

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC	Block Public...
☐	public-web-sub-1b	subnet-0ff59b0fe30bdc66	Available	vpc-052e46fe83a2aa76f dem...	Off
☐	private-app-sub-1a	subnet-07ac55132e6aadbc3	Available	vpc-052e46fe83a2aa76f dem...	Off
☐	private-web-sub-1c	subnet-03e84f41c7b591bff	Available	vpc-052e46fe83a2aa76f dem...	Off
☐	private-DB-sub-1a	subnet-038bdc2207790e850	Available	vpc-052e46fe83a2aa76f dem...	Off
☐	private-app-sub-1b	subnet-07d4d875c492909ba	Available	vpc-052e46fe83a2aa76f dem...	Off
☐	public-web-sub-1a	subnet-0ea60349aceff7f6e	Available	vpc-052e46fe83a2aa76f dem...	Off
☐	private-app-sub-1c	subnet-099e3d6e789cfcdb1	Available	vpc-052e46fe83a2aa76f dem...	Off
☐	private-DB-sub-1c	subnet-08d9add449a889471	Available	vpc-052e46fe83a2aa76f dem...	Off
☐	private-DB-sub-1b	subnet-0e3f29dea7c33bfab	Available	vpc-052e46fe83a2aa76f dem...	Off
☐	private-web-sub-1b	subnet-056f1b3a272015364	Available	vpc-052e46fe83a2aa76f dem...	Off
☐	private-web-sub-1a	subnet-02f8e2a7d7e3f9956	Available	vpc-052e46fe83a2aa76f dem...	Off
☐	public-web-sub-1c	subnet-08eb40900ab4e7203	Available	vpc-052e46fe83a2aa76f dem...	Off

Route tables (6) Info

Last updated less than a minute ago

Actions

Create route table

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-0ebcc31380a9d8283	-	-	Yes	vpc-052e46fe83
☐	-	rtb-0a400dabb2de34017	-	-	Yes	vpc-0a8753599f
☐	private-app-RT	rtb-07abdc03d3aeb8644	3 subnets	-	No	vpc-052e46fe83
☐	private-DB-rt	rtb-0227711e883a77629	3 subnets	-	No	vpc-052e46fe83
☐	private-web-rt	rtb-0185a9adba0eeceab	3 subnets	-	No	vpc-052e46fe83
☐	web-public-routeTB	rtb-003baf40976f94107	3 subnets	-	No	vpc-052e46fe83

You have successfully updated subnet associations for rtb-003baf40976f94107 / web-public-routeTB.

rtb-003baf40976f94107 / web-public-routeTB

Actions

Details Info

Route table ID
rtb-003baf40976f94107

VPC
vpc-052e46fe83a2aa76f | demo-vpc

Main
No

Owner ID
948735414367

Explicit subnet associations
3 subnets

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (3)

Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
public-web-sub-1b	subnet-0ff59b0fe30bdc66	10.0.16.0/20	-
public-web-sub-1a	subnet-0ea60349aceff7f6e	10.0.0.0/24	-
public-web-sub-1c	subnet-08eb40900ab4e7203	10.0.32.0/20	-

NAT gateway nat-0c63840b749dcde19 | demo-NAT-Gateway was created successfully.

nat-0c63840b749dcde19 / demo-NAT-Gateway

Actions

Details

NAT gateway ID

nat-0c63840b749dcde19

NAT gateway ARN

arn:aws:ec2:ap-south-1:948735414367:natgateway/nat-0c63840b749dcde19

VPC

vpc-052e46fe83a2aa76f / demo-vpc

Connectivity type

Public

Primary public IPv4 address

-

Subnet

subnet-0ea60349aceff7f6e / public-web-sub-1a

State

Pending

Primary private IPv4 address

-

Created

Saturday, June 27, 2026 at 10:57:39 GMT+5:30

State message

Info

-

Primary network interface ID

-

Deleted

-

Secondary IPv4 addresses

Monitoring

Tags

Secondary IPv4 addresses

Edit secondary IPv4 address associations

Search

< 1 >

Private IPv4 address

Network interface ID

Status

Failure message

Secondary IPv4 addresses are not available for this nat gateway.

Internet gateways (1/2)

Info

Actions

Create internet gateway

Find internet gateways by attribute or tag

< 1 >

Name

Internet gateway ID

State

VPC ID

Owner

-

igw-0b662c5572695573d

Attached

vpc-0a875359960b93c2d

9487354

demo-IG

igw-00116054c0512e989

Attached

vpc-052e46fe83a2aa76f / demo-vpc

9487354

VPC

Route tables

rtb-003baf40976f94107

Edit routes

Edit routes

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

0.0.0.0/0

local

Internet Gateway

igw-00116054c0512e989

No

CreateRoute

Remove

Add route

Cancel

Preview

Save changes

VPC > Security Groups > Create security group

Create security group

info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

Info

frontend-ALB-SG

Name cannot be edited after creation.

Description

Info

frontend-ALB-SG

VPC

Info

vpc-052e46fe83a2aa76f (demo-vpc)

Inbound rules

Info

Type

Info

Custom TCP

Protocol

Info

TCP

Port range

Info

80

Source

Info

Any...

Description - optional

Info

from internet

0.0.0.0/0

X

Add rule

Delete

VPC > Security Groups > Create security group

Create security group

info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

Info

Frontend-Server-SG

Name cannot be edited after creation.

Description

Info

Frontend-Server-SG

VPC

Info

vpc-052e46fe83a2aa76f (demo-vpc)

Inbound rules

Info

Type

Info

Custom TCP

Protocol

Info

TCP

Port range

Info

80

Source

Info

Custom

Description - optional

Info

from frontend ALB

sg-028e4264a60a3e

X

sg-028e4264a60a3e408

X

Add rule

Delete

VPC > Security Groups > Create security group

Create security group

info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

Info

backend-ALB-SG

Name cannot be edited after creation.

Description

Info

backend-ALB-SG

VPC

Info

vpc-052e46fe83a2aa76f (demo-vpc)

Inbound rules

Info

Type

Info

Custom TCP

Protocol

Info

TCP

Port range

Info

80

Source

Info

Custom

Description - optional

Info

from frontend server

sg-05b8e71241042c

X

sg-05b8e712410420e3a

X

Add rule

Delete

VPC > Security Groups > Create security group

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

backend-server-SG

Name cannot be edited after creation.

Description

backend-server-SG

VPC

vpc-052e46fe83a2aa76f (demo-vpc)

Inbound rules

Type

Custom TCP

Protocol

TCP

Port range

80

Source

Custom

Q

sg-018eb992128ed3

X

sg-

018eb992128ed3dd5

X

Description - optional

from backend ALB

Delete

Add rule

VPC > Security Groups > Create security group

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

Database-SG

Name cannot be edited after creation.

Description

Database-SG

VPC

vpc-052e46fe83a2aa76f (demo-vpc)

Inbound rules

Type

Custom TCP

Protocol

TCP

Port range

3306

Source

Custom

Q

sg-08fe727ce2e486t

X

sg-08fe727ce2e48656d

X

Description - optional

from backend server

Delete

Add rule

Aurora and RDS > Subnet groups

Subnet groups (0)

Filter by subnet group

< 1 >

⚙

Name

Description

Status

VPC

No db subnet groups
You don't have any db subnet groups.
Create DB subnet group

Aurora and RDS

>

Subnet groups

>

Create DB subnet group

Choose the Availability Zone that includes the subnets you want to add.

Choose an availability zone

ap-south-1a Xap-south-1b Xap-south-1c X

Subnets

Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

private-DB-sub-1b
Subnet ID: subnet-0e3f29dea7c33bfab CIDR: 10.0.160.0/20 X

private-DB-sub-1a
Subnet ID: subnet-038bdc2207790e850 CIDR: 10.0.144.0/20 X

private-DB-sub-1c
Subnet ID: subnet-08d9add449a889471 CIDR: 10.0.176.0/20 X

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (3)

Availability zone	Subnet name	Subnet ID	CIDR block
ap-south-1b	private-DB-sub-1b	subnet-0e3f29dea7c33bfab	10.0.160.0/20
ap-south-1a	private-DB-sub-1a	subnet-038bdc2207790e850	10.0.144.0/20
ap-south-1c	private-DB-sub-1c	subnet-08d9add449a889471	10.0.176.0/20

Cancel

Create

database-2

Modify

Actions

Summary

DB identifier

database-2

CPU

-

Status

Creating

Class

db.t4g.micro

Role

Instance

Current activity

Engine

MySQL Community

Region & AZ

ap-south-1c

Recommendations

<

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & back

>

Instance

Configuration

DB instance ID

database-2

Engine version

8.4.8

RDS Extended Support

Disabled

DB name

-

Instance class

Instance class

db.t4g.micro

vCPU

2

RAM

1 GB

Availability

Primary storage

Encryption

Enabled

AWS KMS key

aws/rds

Storage type

General Purpose SSD (gp2)

Storage

20 GiB

Monitoring

Monitoring type

Database Insights - Standard

Performance Insights

Disabled

Enhanced Monitoring

Disabled

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

frontend-ALB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

☐ Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **internet-facing** load balancers only.

Summary

Review and confirm your configurations. [Estimate cost](#) [i](#)

Basic configuration [Edit](#)

Name: frontend-ALB

Scheme: Internet-facing

IP address type: IPv4

Network mapping [i](#) [Edit](#)

VPC: [ypc-052e46fe83a2aa76f](#)

Public IPv4 IPAM pool: -

Availability Zones and subnets:

- ap-south-1a
[subnet-0ea60349aceff7f6e](#)
public-web-sub-1a
- ap-south-1b
[subnet-0ff59b0fe30bdc66](#)
public-web-sub-1b
- ap-south-1c
[subnet-08eb40900ab4e7203](#)
public-web-sub-1c

Security groups [i](#) [Edit](#)

frontend-ALB-SG

[sg-028e4264a60a3e408](#)

Listeners and routing [Edit](#)

HTTP:80 | Forward to 1 target group

Service Integrations [Edit](#)

Amazon CloudFront + AWS Web Application Firewall (WAF): -

AWS WAF: -

AWS Global Accelerator: -

Tags [Edit](#)

-

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

Backend-ALB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme | Info

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type | Info

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ IPv4

- Includes only IPv4 addresses.

☐ Dualstack

- Includes IPv4 and IPv6 addresses.

☐ Dualstack without public IPv4

- Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

Review

Review the load balancer configurations and make changes if needed. After you finish reviewing the configurations, choose **Create load balancer**.

Summary

Review and confirm your configurations. [Estimate cost](#)

Basic configuration [Edit](#)
Name: Backend-ALB
Scheme: Internet-facing
IP address type: IPv4

Network mapping [t](#) [Edit](#) [⚠](#)
VPC: [vpc-052e46fe83a2aa76f](#)
Public IPv4 IPAM pool: -
Availability Zones and subnets:

- ap-south-1a
[subnet-07ac55132e6aadb3](#)
private-app-sub-1a
- ap-south-1b
[subnet-07d4d875c492909ba](#)
private-app-sub-1b
- ap-south-1c
[subnet-099e3d6e789cfcdb1](#)
private-app-sub-1c

Security groups [t](#) [Edit](#)
backend-ALB-SG
[sg-018eb992128ed3dd5](#)

Listeners and routing [Edit](#)
HTTP:80 | [Forward to 1 target group](#)

Load balancers (2) [What's new?](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

< 1 > ⚙

☐	Name	State	Type	Scheme	IP address type	VPC ID	Availability
☐	frontend-ALB	Active	application	Internet-facing	IPv4	vpc-052e46fe83a2aa76f	3 Avail
☐	Backend-ALB	Provisioning..	application	Internet-facing	IPv4	vpc-052e46fe83a2aa76f	3 Avail

Instances (2) [Info](#)

Last updated less than a minute ago

Saved filter sets

[Choose filter set](#)

< 1 > ⚙

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☐	backend-sever-Ami	i-0933bde15fb519e47	Running	t3.micro	Initializing	ap-south-1a
☐	Frontend-server-ami	i-037c8460299c421a0	Running	t3.micro	3/3 checks passec	ap-south-1b

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-22-95 ~]$ sudo dnf install nginx -y
Amazon Linux 2023 Kernel Livepatch repository
Last metadata expiration check: 0:00:01 ago on Sat Jun 27 07:17:24 2026.
Dependencies resolved.

Package Architecture Version Repository Size
Installing:
nginx x86_64 1:1.30.2-1.amzn2023.0.1 amazonlinux 34 k
Installing dependencies:
generic-logos-httpd x86_64 18.0.0-12.amzn2023.0.3 amazonlinux 19 k
gperftools-libs x86_64 2.9.1-1.amzn2023.0.3 amazonlinux 308 k
libunwind x86_64 1.4.0-5.amzn2023.0.3 amazonlinux 66 k
nginx-core x86_64 1:1.30.2-1.amzn2023.0.1 amazonlinux 709 k
nginx-filesystem noarch 1:1.30.2-1.amzn2023.0.1 amazonlinux 10 k
nginx-mimetypes noarch 2.1.49-3.amzn2023.0.3 amazonlinux 21 k

Transaction Summary
Install 7 Packages
```

```
Complete!
[ec2-user@ip-10-0-22-95 ~]$ sudo dnf install git -y
Last metadata expiration check: 0:00:31 ago on Sat Jun 27 07:17:24 2026.
Dependencies resolved.

Package Architecture Version Repository Size
Installing:
git x86_64 2.50.1-1.amzn2023.0.1 amazonlinux 53 k
Installing dependencies:
git-core x86_64 2.50.1-1.amzn2023.0.1 amazonlinux 4.9 M
git-core-doc noarch 2.50.1-1.amzn2023.0.1 amazonlinux 2.8 M
perl-Error noarch 1:0.17030-2.amzn2023.0.1 amazonlinux 42 k
perl-File-Find noarch 1.37-477.amzn2023.0.9 amazonlinux 26 k
perl-Git noarch 2.50.1-1.amzn2023.0.1 amazonlinux 41 k
perl-TermReadKey x86_64 2.38-9.amzn2023.0.3 amazonlinux 36 k
perl-lib x86_64 0.65-477.amzn2023.0.9 amazonlinux 15 k

Transaction Summary
Install 8 Packages
```

```
Complete!
[ec2-user@ip-10-0-22-95 ~]$ sudo vi /etc/nginx/nginx.conf
[ec2-user@ip-10-0-22-95 ~]$ sudo vi /etc/nginx/nginx.conf
[ec2-user@ip-10-0-22-95 ~]$ sudo vim /etc/nginx/nginx.conf
[ec2-user@ip-10-0-22-95 ~]$ sudo vim /etc/nginx/nginx.conf
[ec2-user@ip-10-0-22-95 ~]$ sudo vim /etc/nginx/nginx.conf
[ec2-user@ip-10-0-22-95 ~]$
```

```
# Load modular configuration files from the /etc/nginx/conf.d directory.
# See http://nginx.org/en/docs/nginx_core_module.html#include
# for more information.
include /etc/nginx/conf.d/*.conf;

server {
    listen 80;
    server_name localhost;
    root /usr/share/nginx/html;
    index index.html;

    # Static content
    location ~* \.(jpg|jpeg|png|gif|ico|css|js)$ {
        expires 30d;
        add_header Cache-Control "public, no-transform";
    }

    # API requests forwarding to PHP backend
    location /api/ {
        proxy_pass http://update-me;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }

    # Main fallback handler
    location / {
        try_files $uri $uri/ /index.html;
    }
}

# server {
#     listen 80;
#     listen [::]:80;
```


Instances (1/2) Info

Last updated 18 minutes ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

1

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☒	backend-sever-Ami	i-0933bde15fb519e47	Running	t3.micro	3/3 checks passed	ap-south-1a
☐	Frontend-server-ami	i-037c8460299c421a0	Running	t3.micro	3/3 checks passed	ap-south-1b

To prepare the LAMP server

1. Connect to your instance. For more information, see [Connecting to AL2023 instances](#).
2. To ensure that all of your software packages are up to date, perform a quick software update on your instance. This process might take a few minutes, but it is important to make sure that you have the latest security updates and bug fixes.

The `-y` option installs the updates without asking for confirmation. If you would like to examine the updates before installing, you can omit this option.

```
[ec2-user ~]$ sudo dnf upgrade -y
```

3. Install the latest versions of Apache web server and PHP packages for AL2023.

```
[ec2-user ~]$ sudo dnf install -y httpd wget php-fpm php-mysql php-j
```

4. Install the MariaDB software packages. Use the `dnf install` command to install multiple software packages and all related dependencies at the same time.

```
[ec2-user ~]$ sudo dnf install mariadb105-server
```

You can view the current versions of these packages using the following command:

```
[ec2-user ~]$ sudo dnf info package_name
```

5. Start the Apache web server.

```
[ec2-user ~]$ sudo systemctl start httpd
```



6. Use the **systemctl** command to configure the Apache web server to start at each system boot.

```
[ec2-user ~]$ sudo systemctl enable httpd
```



You can verify that **httpd** is on by running the following command:

```
[ec2-user ~]$ sudo systemctl is-enabled httpd
```



3. Change the group ownership of `/var/www` and its contents to the `apache` group.

```
[ec2-user ~]$ sudo chown -R ec2-user:apache /var/www
```



4. To add group write permissions and to set the group ID on future subdirectories, change the directory permissions of `/var/www` and its subdirectories.

```
[ec2-user ~]$ sudo chmod 2775 /var/www && find /var/www -type d -exec
```



5. To add group write permissions, recursively change the file permissions of `/var/www` and its subdirectories:

```
[ec2-user ~]$ find /var/www -type f -exec sudo chmod 0664 {} \;
```



```
/m/'
Last login: Sat Jun 27 07:32:16 2026 from 13.233.177.4
[ec2-user@ip-10-0-0-58 ~]$ mysql -h database-2.cliiysg6siwc.ap-south-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 71
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> █
```

```
Code Blame 18 lines (15 loc) · 510 Bytes
1  -- database_setup.sql
2
3  -- Create database
4  CREATE DATABASE IF NOT EXISTS hello_world;
5  USE hello_world;
6
7  -- Create messages table
8  CREATE TABLE IF NOT EXISTS messages (
9      id INT AUTO_INCREMENT PRIMARY KEY,
10     message VARCHAR(255) NOT NULL,
11     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
12 );
13
14 -- Insert some initial data
15 INSERT INTO messages (message) VALUES
16 ('Hello from the database!'),
17 ('Welcome to our three-tier architecture demo'),
18 ('This is a simple example showing frontend, backend, and database');
```

```
MySQL [(none)]> -- database_setup.sql
MySQL [(none)]>
MySQL [(none)]> -- Create database
MySQL [(none)]> CREATE DATABASE IF NOT EXISTS hello_world;
Query OK, 1 row affected (0.051 sec)

MySQL [(none)]> USE hello_world;
Database changed
MySQL [hello_world]>
MySQL [hello_world]> -- Create messages table
MySQL [hello_world]> CREATE TABLE IF NOT EXISTS messages (
->     id INT AUTO_INCREMENT PRIMARY KEY,
->     message VARCHAR(255) NOT NULL,
->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.057 sec)

MySQL [hello_world]>
MySQL [hello_world]> -- Insert some initial data
MySQL [hello_world]> INSERT INTO messages (message) VALUES
-> ('Hello from the database!'),
-> ('Welcome to our three-tier architecture demo'),
-> ('This is a simple example showing frontend, backend, and database');
Query OK, 3 rows affected (0.012 sec)
Records: 3 Duplicates: 0 Warnings: 0

MySQL [hello_world]> █
```

Owned by me ▼ Find AMI by attribute or tag						1 ▼ ⚙️	
☐	Name 🔗 ▼	AMI name ▼	AMI ID ▼	Source ▼	Owner ▼		
☐		backend-sever-Ami	ami-0ce3f63da7db79574	948735414367/backend-sever-Ami	948735414367		
☐		frontend-Server-ami	ami-028419b7d6ce7024b	948735414367/frontend-Server-ami	948735414367		

EC2 > Launch templates > Create launch template

Don't include in launch template

Allow tags in metadata Info

Don't include in launch template ▼

User data - optional Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
sed -i 's/update-me/Backend-ALB-1367272452.ap-south-1.elb.amazonaws.com/g'
/etc/nginx/nginx.conf
sudo systemctl start nginx
cd /usr/share/nginx/html
sudo git clone https://github.com/ajitinamdar-tech/three-tier-architecture-aws.git
mv /usr/share/nginx/html/three-tier-architecture-aws/frontend/*
/usr/share/nginx/html/
sudo rm -rf /usr/share/nginx/html/three-tier-architecture-aws
```

☐ User data has already been base64 encoded

▼ Summary

Software Image (AMI)

frontend-Server-ami
ami-028419b7d6ce7024b

Virtual server type (instance type)

-

Firewall (security group)

Frontend-Server-SG

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Create launch template

EC2 > Launch templates > Create launch template

Don't include in launch template

Allow tags in metadata Info

Don't include in launch template ▼

User data - optional Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
sudo systemctl start httpd
cd /var/www/html
sudo git clone https://github.com/ajitinamdar-tech/three-tier-architecture-aws.git
sudo mkdir api
sudo mv /var/www/html/three-tier-architecture-aws/backend/api/*
/var/www/html/api/
sudo rm -rf /var/www/html/three-tier-architecture-aws
sed -i 's/update-me-host/database-2.cliysg6siwc.ap-south-1.rds.amazonaws.com/g'
/var/www/html/api/db_connection.php
sed -i 's/update-me-username/admin/g' /var/www/html/api/db_connection.php
sed -i 's/update-me-password/hasrat123/g' /var/www/html/api/db_connection.php
```

☐ User data has already been base64 encoded

▼ Summary

Software Image (AMI)

backend-sever-Ami
ami-0ce3f63da7db79574

Virtual server type (instance type)

-

Firewall (security group)

backend-server-SG

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Create launch template

Backend-server-ASG, 1 Scaling policy created successfully

Auto Scaling groups (2) Info

Last updated less than a minute ago

Launch configurations

Launch templates

Actions

Create Auto Scaling group

Search your Auto Scaling groups

< 1 >

☐	Name	Status - new	Launch template/configuration	Instances	Instance health - new	Desired capacity	Min
☐	Backend-server-ASG	At desired capacity	Backend-server-LT Version Default	2	2/2 Healthy	2	2
☐	frontend-server-ASG	At desired capacity	frontend-server-LT Version Default	2	2/2 Healthy	2	2

frontend-server-ASG

frontend-server-ASG Capacity overview

Edit

arn:aws:autoscaling:ap-south-1:948735414367:autoScalingGroup:df907bfd-220e-4a4e-8aaf-fb28c5a193f5:autoScalingGroupName/frontend-server-ASG

Desired capacity2

Scaling limits2 - 4

Desired capacity typeUnits (number of instances)

StatusAt desired capacity

Date createdSat Jun 27 2026 14:52:08 GMT+0530 (India Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Launch template

Edit

Launch template

lt-0256e359f8c699172

frontend-server-LT

Version

Default

Description

-

AMI ID

ami-028419b7d6ce7024b

Security groups

-

Storage (volumes)

-

Instance type

-

Security group IDs

sg-05b8e712410420e3a

Key pair name

3tier

Owner

arn:aws:iam::948735414367:user/Hasrat-IAM

Create time

Sat Jun 27 2026 14:38:41 GMT+0530 (India Standard Time)

Request Spot Instances

No

Backend-server-ASG

Backend-server-ASG Capacity overview

Edit

arn:aws:autoscaling:ap-south-1:948735414367:autoScalingGroup:3c1a95c3-c2c0-468b-a55f-2cf47193658e:autoScalingGroupName/Backend-server-ASG

Desired capacity2

Scaling limits2 - 4

Desired capacity typeUnits (number of instances)

StatusAt desired capacity

Date createdSat Jun 27 2026 14:55:39 GMT+0530 (India Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Launch template

Edit

Launch template

lt-08cb33a05ccadd9b4

Backend-server-LT

Version

Default

Description

-

AMI ID

ami-0ce3f63da7db79574

Security groups

-

Storage (volumes)

-

Instance type

-

Security group IDs

sg-08fe727ce2e48656d

Key pair name

3tier

Owner

arn:aws:iam::948735414367:user/Hasrat-IAM

Create time

Sat Jun 27 2026 14:44:28 GMT+0530 (India Standard Time)

Request Spot Instances

No

Instances (6) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

< 1 >

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Publ
☐	Backend-Server	i-05a386b9cfe7ab128	Running	t3.micro	Initializing	ap-south-1c	-
☐	Frontend-server	i-03c754b8b338c5393	Running	t3.micro	Initializing	ap-south-1c	-
☐	backend-sever-Ami	i-0933bde15fb519e47	Stopped	t3.micro	-	ap-south-1a	-
☐	Frontend-server	i-0a17624a167d681c9	Running	t3.micro	Initializing	ap-south-1a	-
☐	Frontend-server-ami	i-037c8460299c421a0	Stopped	t3.micro	-	ap-south-1b	-
☐	Backend-Server	i-0ea6b27cf89d64453	Running	t3.micro	Initializing	ap-south-1b	-

Load balancers (1/2) What's new?

Actions

Create load balancer

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

< 1 >

☐	Name	IPs	Security groups	DNS name	ARN	Date created
☒	frontend-ALB	1es	sg-028e4264a60a3e40...	frontend-ALB-763675877.a...	ARN	June 27, 2026, 12:20 (UTC+05:...
☐	Backend-ALB	1es	sg-018eb992128ed3dd...	Backend-ALB-1367272452....	ARN	June 27, 2026, 12:28 (UTC+05:...

Tech with Ajit

Basic Three-Tier Architecture Example

Message from Database:

Hello from the database! (2025-04-29 20:58:29)

Welcome to our three-tier architecture demo (2025-04-29 20:58:29)

This is a simple example showing frontend, backend, and database (2025-04-29 20:58:29)

Add New Message

New Message:

Save Message

Tech with Ajit

Basic Three-Tier Architecture Example

Message from Database:

- Demo from youtube (2025-04-29 21:17:25)
- Hello from the database! (2025-04-29 20:58:29)
- Welcome to our three-tier architecture demo (2025-04-29 20:58:29)
- This is a simple example showing frontend, backend, and database (2025-04-29 20:58:29)

Add New Message

New Message:

Save Message